4. Electromagnets

Ørsted's accident, repeated

A current makes a magnetic field

Lesson 4 · about 75 minutes · everyone builds; battery power only

What you need

- Enamelled magnet wire, 26–30 AWG, about 10 m
- Large iron nail or bolt (100 mm)
- AA cells and holder; a 9 V battery
- Compass
- Paper clips
- Multimeter
- Sandpaper, tape

Predict first

A compass sits under a straight wire that runs north–south. You connect the battery. Which way does the needle turn — and what happens if you reverse the battery? Then: an electromagnet with 50 turns lifts 8 paper clips. How many will 100 turns lift? Commit to a number.

Do it

1. **Ørsted's experiment.** Lay a straight wire over a compass, running the same way the needle points. Touch it briefly across a AA cell. The needle swings across. Reverse the connections — it swings the other way. In 1820 this was the first evidence that electricity and magnetism are one subject. *Only touch it briefly:* a bare wire across a cell is a short circuit and gets hot.
2. **Find the direction.** Move the compass above, below and beside the wire. The field circles *around* the wire. Point your right thumb along the current and your fingers curl the way the field goes.
3. **Wind a coil.** Wind 50 turns of magnet wire neatly along the nail, all the same direction, counting aloud. Leave 15 cm tails, sand the enamel off the ends — enamel is insulation and the circuit will not work through it.
4. **Lift things.** Connect to the AA holder and pick up paper clips. Count them. Disconnect: they fall.
5. **Double it.** Wind 50 more turns on top, same direction, so you now have 100. Count clips again. Compare with your prediction.
6. **Take the iron out.** Slide the nail out and try lifting with the air-cored coil alone. Then put it back.
7. **Measure the current.** Put the meter in series and record it for the 50-turn and 100-turn coils. The 100-turn coil has twice the wire, so twice the resistance, so *half* the current — which complicates your prediction in an interesting way.

What it means

Current makes field	Not "moving magnets make fields" — moving *charge* makes a magnetic field. A bar magnet works because of circulating currents inside its atoms.
Turns concentrate it	Each turn adds its field to the others. The strength depends on the **ampere-turns**: current multiplied by number of turns. That is why your measurement mattered — doubling turns while halving current is roughly a wash, and the honest answer to the prediction is "about the same, unless I also change the voltage".
Iron multiplies	An iron core can multiply the field by hundreds. The iron's own atomic magnets line up with yours and add to it. This is why transformers have iron cores — and why a Tesla coil, which works at radio frequency where iron cannot keep up, has an *air* core.
Switching it off	A permanent magnet is always on. This one is a magnet you can control, which is the entire basis of motors, relays, loudspeakers and transformers.

Break it

Wind it backwards	Wind the second 50 turns the *opposite* way. Now what happens? Predict before you connect.
More voltage	Try 9 V instead of 3 V on the same coil. Measure the current. Feel the coil after twenty seconds — and think about where that energy is going.
Aluminium core	Try a core of aluminium or copper instead of iron. Nothing happens. Only a few metals are ferromagnetic, which is a fact about atoms and not about metals in general.

The link to the coil

The primary coil of the Tesla coil is exactly this: a few turns of heavy conductor making a magnetic field. The secondary is the same idea with about a thousand turns of thin wire. Neither has an iron core, because at the coil's operating frequency — hundreds of thousands of cycles per second — iron cannot flip fast enough and would simply heat up.

> **This lesson is low voltage.** Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.